Smart Speed Monitoring System for Road Safety

Requirement Analysis

Prepared by

Ayush Kishor Dharmik

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1.0 Introduction

Road accidents caused by over-speeding remain one of the most significant contributors to fatalities on both urban and rural road networks. Conventional speed enforcement relies heavily on manual traffic policing or fixed radar-based systems, both of which are expensive to deploy at scale and are largely reactive rather than preventive in nature. The Smart Speed Monitoring System for Road Safety has been conceived as a low-cost, embedded-systems-based alternative that can be deployed at accident-prone stretches, school zones, and residential areas to provide real-time vehicle speed estimation and driver feedback.

This project uses infrared (IR) sensor modules positioned at a known fixed distance along the direction of vehicle travel. By measuring the time interval between successive sensor triggers, the embedded controller computes the instantaneous speed of a passing vehicle and displays it on an OLED module. Should the computed speed exceed a pre-configured threshold, an audible buzzer alert is triggered to warn the driver and nearby pedestrians. The system is designed around the Arduino Uno platform, chosen for its wide availability, low cost, and suitability for rapid prototyping within an academic and internship setting.

This document presents the complete requirement analysis carried out during the Embedded Systems Internship at Arnita Infotech. It captures the functional and non-functional requirements, the hardware and software components selected, the constraints under which the system must operate, and the assumptions and limitations that shaped the design decisions. The analysis presented here forms the foundation for the subsequent design, implementation, and testing phases of the project.

2.0 Objective

The primary objective of this project is to design and prototype an embedded system capable of detecting the speed of a moving vehicle using low-cost infrared sensors and communicating that information to the driver in real time through a visual display and an audible alert. The system is intended to serve as a proof-of-concept for low-cost speed enforcement and driver-awareness solutions that can be installed at locations where conventional radar-based enforcement is not economically viable.

Specific objectives of the project include accurately measuring vehicle speed using a two-sensor time-of-flight approach, displaying the computed speed on an OLED display in a legible format, triggering a buzzer alert whenever a configurable speed threshold is exceeded, and ensuring that the entire system can be built and validated using components readily available in an academic laboratory. A further objective is to structure the firmware and hardware design in a manner that permits future extension toward IoT connectivity, data logging, and more advanced vehicle-detection technologies without requiring a complete redesign.

3.0 Problem Statement

Over-speeding continues to be one of the leading causes of road traffic accidents, particularly in school zones, residential colonies, and sharp-curve sections of highways where reaction time for both drivers and pedestrians is limited. Existing enforcement mechanisms such as fixed radar guns, ANPR-integrated speed cameras, and manned checkpoints require significant capital investment, ongoing maintenance, and trained personnel, making them impractical for widespread deployment across every vulnerable stretch of road, especially in developing regions and semi-urban areas.

There is, therefore, a clear need for a compact, low-cost, and easily deployable speed-monitoring device that can operate autonomously, provide instantaneous feedback to drivers, and be installed without requiring specialized civil infrastructure. This project addresses that need by proposing an embedded solution built around commonly available microcontroller and sensor components, capable of estimating vehicle speed with acceptable accuracy for the purpose of driver alerting rather than legal enforcement.

4.0 Scope of the Project

The scope of this project covers the design, assembly, and testing of a prototype speed-monitoring unit built on the Arduino Uno platform. The system measures the speed of a single vehicle at a time as it passes between two IR sensor modules positioned at a fixed, known separation distance, computes the corresponding speed, and displays the result on an OLED screen while triggering a buzzer when the speed exceeds a configured limit.

The current scope is limited to a single-lane detection scenario under controlled test conditions and does not include multi-vehicle tracking, number plate recognition, or persistent data logging, all of which are identified as future extensions rather than deliverables of the present internship phase. The project scope explicitly includes requirement analysis, hardware selection, circuit design, firmware development, and functional testing using the Wokwi simulator prior to physical hardware validation.

Requirement Analysis

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Hardware Selection

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Circuit Design & Simulation (Wokwi)

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Firmware Development

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Hardware Assembly & Integration

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Testing & Validation

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Documentation

Figure 1: Project Workflow

5.0 Functional Requirements

The system shall detect the presence of a vehicle passing over each of the two IR sensor modules and record the corresponding timestamps using the microcontroller's internal timer. It shall compute the elapsed time between the two detection events and use the known physical distance between the sensors to calculate the vehicle's speed using the standard distance-over-time relationship, converting the result into kilometres per hour for display.

The system shall continuously display the most recently computed speed value on the OLED display, along with a static status indicator distinguishing normal operation from an over-speed condition. Whenever the computed speed exceeds a pre-defined threshold value stored in the firmware, the system shall activate the buzzer for a fixed duration to alert the driver, and shall reset itself to a ready state after each vehicle detection cycle so that it is prepared to measure the next passing vehicle without manual intervention.

6.0 Non-Functional Requirements

Beyond its core functional behaviour, the system is expected to satisfy several non-functional qualities that determine its practical usability. It must respond to a passing vehicle and update the display within a time frame short enough to be perceived as real time by an observer, and it must produce consistent, repeatable speed readings across multiple test runs under similar conditions.

The system should be simple enough to install and calibrate without specialized tools, and its firmware should be structured in a modular fashion so that individual components such as the sensor-reading routine, the speed-calculation routine, and the display-update routine can be maintained or replaced independently. Power consumption should remain low enough to permit operation from a standard USB power source or a small battery pack, and the overall bill of materials should remain low-cost enough to justify deployment at multiple locations.

7.0 Hardware Requirements

The hardware architecture of the Smart Speed Monitoring System is centred around the Arduino Uno microcontroller board, which handles sensor input processing, speed computation, and output control. Two IR sensor modules are positioned at a fixed known distance apart along the vehicle's path of travel and are used to mark the entry and exit timestamps required for speed calculation. An OLED display communicating over the I^2C protocol presents the computed speed and system status to an observer, while an active buzzer provides an audible over-speed alert.

A breadboard and a set of jumper wires are used to interconnect all components during the prototyping phase, and a standard USB cable supplies power to the Arduino Uno from a computer or a USB power adapter. The complete hardware bill of materials is summarised in the table below.

Component	Quantity	Purpose
Arduino Uno	1	Central microcontroller unit; reads sensor inputs, computes vehicle speed, and drives the display and buzzer outputs.
IR Sensor Module	2	Detects vehicle passage at two fixed points to provide the timestamps used for speed calculation.
OLED Display (I^2C)	1	Displays the computed vehicle speed and system status in real time.
Active Buzzer	1	Generates an audible alert when the computed speed exceeds the configured threshold.
Breadboard	1	Provides a solderless platform for interconnecting components during prototyping.
Jumper Wires	As required	Provide electrical connections between the Arduino Uno, sensors, display, and buzzer.
USB Cable	1	Supplies power to the Arduino Uno and enables firmware upload from a computer.

8.0 Software Requirements

The firmware for this project is developed using the Arduino IDE, which provides the compiler toolchain and serial upload mechanism required to program the Arduino Uno. The application logic itself is written in Embedded C/C++, following the Arduino programming conventions for setup and loop execution. Communication with the OLED display is handled through the Wire library for I^2C communication together with a dedicated OLED display library that provides text-rendering primitives.

Version control and source-code management for the project are maintained using GitHub, allowing incremental development and providing a record of design changes throughout the internship. Prior to physical hardware assembly, the circuit and firmware logic are validated using the Wokwi online simulator, which allows the sensor-triggering behaviour and display output to be verified in a virtual environment, reducing the risk of hardware damage during the debugging phase.

Software / Tool	Description
Arduino IDE	Development environment used to write, compile, and upload firmware to the Arduino Uno.
Embedded C/C++	Programming language used to implement the sensor-reading, speed-calculation, and control logic.
Wire Library	Provides I²C communication between the Arduino Uno and the OLED display module.
OLED Library	Supplies text-rendering and display-control functions for the I²C OLED module.
GitHub	Used for source-code version control and change tracking throughout development.
Wokwi Simulator	Used to simulate and validate circuit behaviour and firmware logic prior to physical assembly.

9.0 System Inputs

The primary inputs to the system are the digital trigger signals generated by the two IR sensor modules whenever a vehicle passes in front of each sensor's detection beam. These digital signals are read by the Arduino Uno's input pins and are used to timestamp the moment of vehicle entry and vehicle exit relative to the fixed sensor spacing.

In addition to the real-time sensor signals, the system also relies on two configuration inputs that are set within the firmware prior to deployment: the physical distance between the two IR sensors, and the speed threshold above which the over-speed alert should be triggered. These configuration values can be adjusted in the source code to suit the specific installation location and the applicable speed limit.

IR Sensor 1 + IR Sensor 2 (Vehicle Detection)

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Arduino Uno (Microcontroller Unit)

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OLED Display + Active Buzzer (Driver Alert)

Figure 2: System Block Diagram

10.0 System Outputs

The system produces two primary outputs. The first is a visual output presented on the OLED display, which shows the most recently calculated vehicle speed in kilometres per hour along with a simple status message indicating whether the reading is within the normal range or has exceeded the configured threshold. The display refreshes automatically after each vehicle detection event.

The second output is an audible alert generated by the active buzzer, which sounds for a short, fixed duration whenever an over-speed condition is detected. This dual visual-and-audible output ensures that both the approaching driver and any nearby pedestrian or traffic observer are made aware of a speed violation without requiring continuous monitoring of the display.

Sensing Layer — IR Sensor Inputs

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Processing Layer — Speed Computation & Threshold Check

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Actuation Layer — OLED Display & Buzzer Output

Figure 3: Embedded System Design Flow (powered throughout by a regulated 5V USB supply)

11.0 Performance Requirements

Performance requirements for the Smart Speed Monitoring System have been defined across six dimensions that directly affect the usability and reliability of the device in a real-world roadside setting.

Response Time: The system must compute and display the vehicle speed within a fraction of a second of the vehicle clearing the second IR sensor, ensuring that the feedback is perceived as immediate by the driver.

Accuracy: The computed speed value must remain within an acceptable margin of the vehicle's true speed under the test conditions used, with the primary sources of error being sensor response latency and imprecision in the measured sensor spacing.

Reliability: The system must consistently detect vehicle passage across repeated test cycles without missed triggers or false detections caused by ambient IR interference or sensor misalignment.

Power Consumption: The system must operate reliably from a standard 5-volt USB supply, with total current draw remaining low enough to permit battery-powered operation for extended field trials.

Maintainability: The modular firmware structure must allow individual routines, such as the speed-calculation logic or the display-formatting logic, to be modified or replaced without affecting the rest of the codebase.

Scalability: The hardware and firmware design must accommodate future extension, such as the addition of communication modules or additional sensors, without requiring a fundamental redesign of the existing architecture.

12.0 Cost Constraints

A central design goal of this project is affordability, since the intended deployment scenario involves installation at numerous locations rather than a single high-value site. The complete bill of materials, comprising the Arduino Uno, two IR sensor modules, an OLED display, a buzzer, a breadboard, jumper wires, and a USB cable, has been selected to keep the per-unit prototype cost significantly lower than that of a commercial radar-based speed-enforcement system.

This cost constraint directly influenced the choice of IR sensors over more expensive alternatives such as ultrasonic or laser-ranging modules, and it also motivated the decision to defer higher-cost future enhancements such as GSM modules, cameras, and cloud connectivity to a later phase of the project rather than including them in the initial prototype.

13.0 Size Constraints

The physical footprint of the system has been kept compact to allow installation on roadside poles, barricades, or temporary mounting structures without requiring dedicated civil work. The Arduino Uno, OLED display, and buzzer are small enough to be housed within a single weatherproof enclosure, while the two IR sensor modules are mounted separately at the required detection spacing along the roadside.

Because the sensor spacing directly determines the achievable speed-measurement resolution, there is an inherent trade-off between the compactness of the installation and the accuracy of the speed computation; a larger separation generally improves timing resolution but increases the physical space required for installation, and this trade-off has been considered during the requirement-analysis phase.

14.0 Environmental Constraints

Because the system is intended for outdoor roadside deployment, several environmental factors have been identified as constraints that influence both the hardware selection and the enclosure design for the final product.

Rain: Direct exposure to rain can damage the exposed electronics and interfere with the IR sensor's optical detection path; the prototype therefore assumes a weatherproof enclosure for any outdoor field deployment beyond laboratory testing.

Dust: Accumulated dust on the IR sensor lens can reduce detection sensitivity over time, making periodic cleaning or the use of a protective transparent cover necessary for sustained field operation.

Fog: Dense fog can scatter the infrared beam and reduce the effective detection range of the IR sensors, potentially leading to missed or delayed triggers under poor visibility conditions.

Sunlight: Strong ambient sunlight, particularly direct sunlight falling on the sensor, can introduce infrared interference that affects the reliability of the detection threshold used by the IR modules.

Temperature: Extreme ambient temperatures can affect the electrical characteristics of the sensor components and the display, and the system's operating range should be verified against the manufacturer specifications of each component.

IR Interference: Other nearby infrared sources, including sunlight, vehicle headlights, and adjacent IR-based devices, can introduce false triggers or missed detections, and shielding or sensor placement adjustments may be required to mitigate this effect.

15.0 Operating Conditions

The prototype has been designed for evaluation under standard indoor laboratory conditions using the Wokwi simulator and controlled physical test setups, with ambient temperature and lighting conditions typical of an indoor engineering laboratory. For eventual field deployment, the system would need to be validated across the full range of expected outdoor operating conditions, including varying temperature, humidity, and ambient light levels described in the environmental constraints section.

The system is expected to operate continuously once powered, remaining in a ready state between vehicle detections and returning to that ready state automatically after each measurement cycle, without requiring manual reset or operator intervention during normal operation.

16.0 Assumptions

The requirement analysis and subsequent design of this project rest on several assumptions that were necessary to scope the work within the available time and resources of the internship. It is assumed that vehicles pass through the detection zone in a single file, one at a time, and that the two IR sensors are aligned precisely enough that both are triggered by the same vehicle without ambiguity.

It is further assumed that the distance between the two sensors is measured accurately and remains fixed for the duration of operation, that the sensors are mounted at a height and angle appropriate for detecting the passage of standard passenger vehicles, and that the ambient lighting conditions during testing do not introduce significant IR interference. Power supply to the Arduino Uno is assumed to be stable and uninterrupted throughout each test session.

17.0 Limitations

The current prototype is subject to several limitations that define the boundary of what has been achieved during this internship phase. The system can reliably process only one vehicle at a time; if a second vehicle enters the detection zone before the first measurement cycle completes, the resulting reading may be inaccurate.

The two-point IR timing method used for speed estimation is inherently less precise than radar or LiDAR-based measurement techniques, and its accuracy is sensitive to sensor alignment and mounting precision. The system does not currently include any means of identifying or logging individual vehicles, nor does it provide remote notification or long-term data storage, all of which have been identified as candidates for future development rather than limitations to be resolved within the current scope.

18.0 Future Scope

While the current prototype satisfies the core objective of real-time speed detection and driver alerting, several enhancements have been identified that would significantly extend its practical value in an actual road-safety deployment.

GSM SMS Alerts: Integrating a GSM module would allow the system to send an SMS notification to a designated traffic authority or control room whenever an over-speed event is detected, enabling remote awareness without requiring an on-site observer.

RTC Date & Time Logging: Adding a real-time clock module would allow each detection event to be timestamped accurately, supporting later analysis of speed-violation patterns by time of day.

Traffic Statistics: With persistent logging in place, the system could be extended to compute aggregate statistics such as average speed, peak-hour traffic volume, and violation frequency over a given period.

ESP32 IoT Integration: Replacing or supplementing the Arduino Uno with an ESP32 module would introduce Wi-Fi connectivity, allowing the system to transmit data to a remote server in real time.

Cloud Dashboard: A cloud-based dashboard could aggregate data from multiple deployed units, giving traffic authorities a centralised view of speed-violation trends across several locations simultaneously.

Camera-based Vehicle Detection: Incorporating a camera module would allow visual confirmation of detected vehicles and would open the possibility of vehicle classification alongside speed measurement.

AI-based Multi-Vehicle Tracking: Machine-learning-based image processing could enable the system to track and measure the speed of multiple vehicles simultaneously, overcoming the single-vehicle limitation of the current design.

ANPR (Automatic Number Plate Recognition): Combined with camera-based detection, automatic number plate recognition would allow violations to be linked to specific vehicles for enforcement purposes.

Radar/LiDAR Integration: Replacing the IR sensor pair with a radar or LiDAR module would substantially improve measurement accuracy and range, particularly under adverse weather conditions identified in the environmental constraints section.

19.0 Conclusion

The Smart Speed Monitoring System for Road Safety demonstrates that a functional, low-cost vehicle speed-detection and driver-alerting device can be built using widely available embedded systems components, without reliance on expensive radar or camera-based infrastructure. Through the requirement analysis presented in this document, the project has established a clear set of functional and non-functional requirements, identified an appropriate hardware and software architecture, and defined the constraints and assumptions that will guide the design and implementation phases that follow.

The prototype, centred on an Arduino Uno with dual IR sensors, an OLED display, and a buzzer, provides a solid foundation for a device capable of measuring vehicle speed and alerting drivers in real time. While the current implementation has known limitations, particularly around single-vehicle detection and measurement precision, the future-scope enhancements outlined in this document, ranging from GSM alerting to full IoT and camera-based integration, present a credible roadmap for evolving this prototype into a deployable road-safety solution. This requirement analysis, completed as part of the Embedded Systems

Internship at Arnita Infotech, provides the necessary groundwork for the subsequent design and development stages of the project.